From Bed Rest to Mars

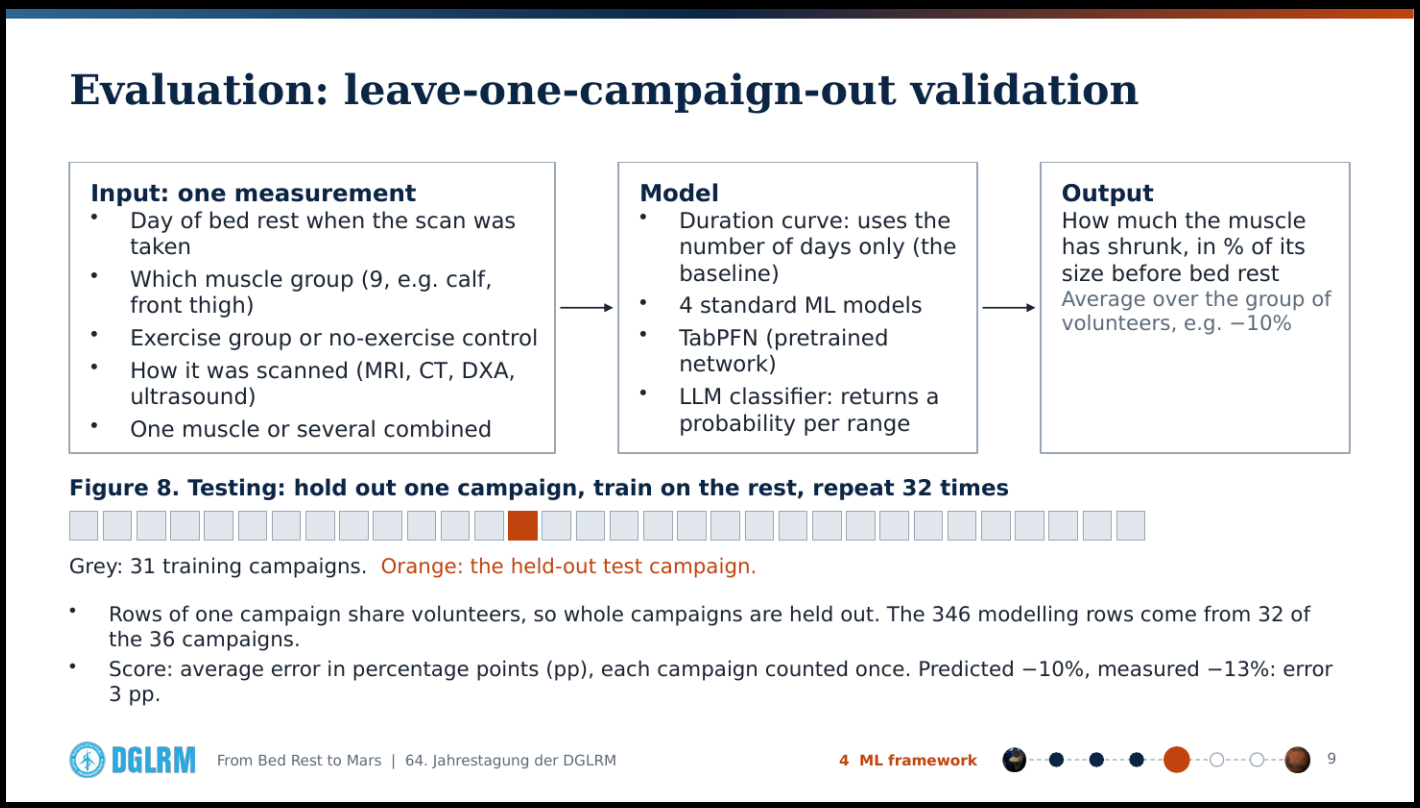
Practice script, Milad's part

Slides 9 to 14: about six minutes. Backup slides 15 to 21 are for questions only.

Amber italic lines are only for answering questions; do not read them out in the talk.

Slide 9 · Evaluation: leave-one-campaign-out validation

About one minute



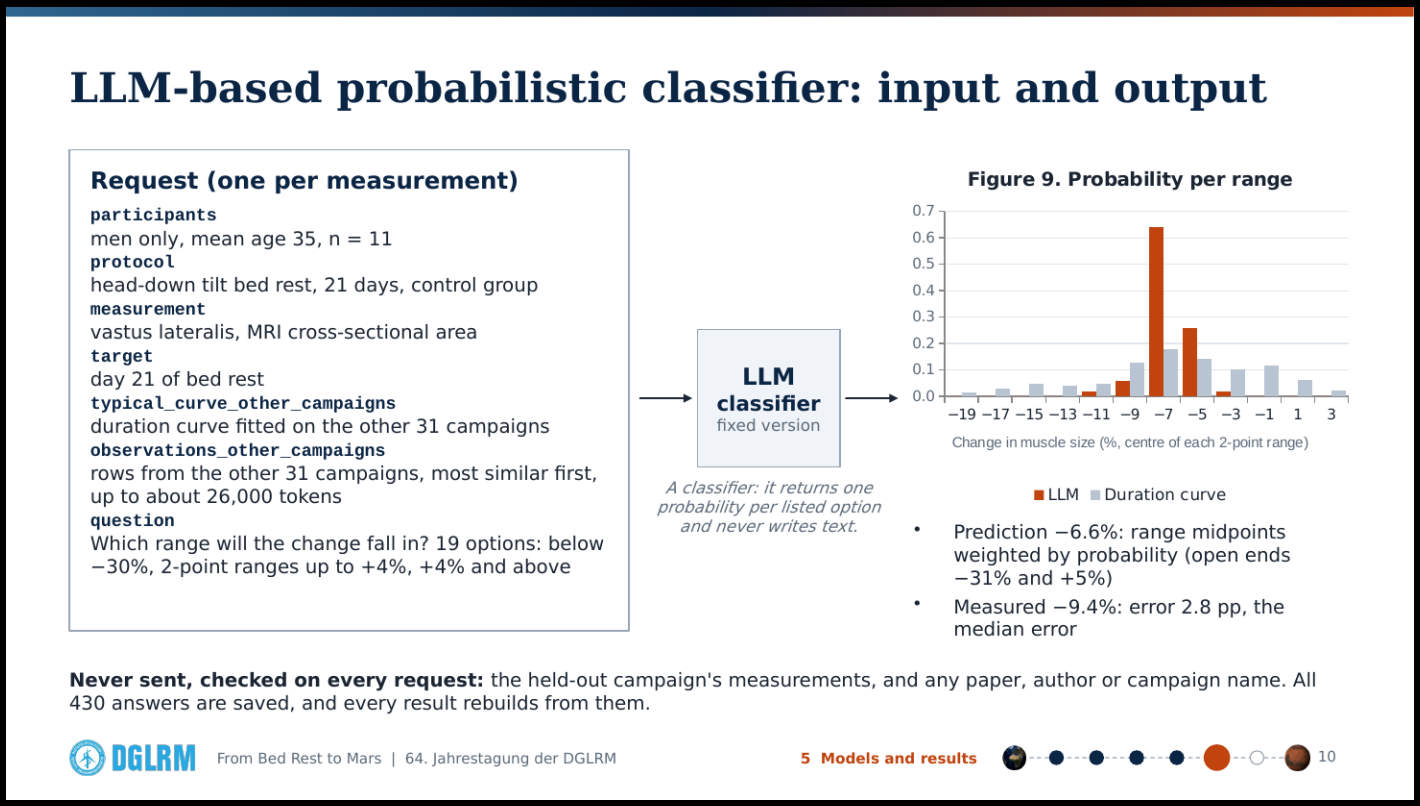
Each row in the dataset is one measurement: the mean change of one muscle, in one group of volunteers, on one day of bed rest. The inputs are the day of the scan, the muscle group, whether the group received a countermeasure, the imaging method, and whether the value refers to one muscle or several combined. The output is the percentage change in muscle size from baseline.

The 346 modelling rows come from 32 independent campaigns. Rows from the same campaign share volunteers, so they are not independent. We therefore used leave-one-campaign-out validation. We trained on 31 campaigns, predicted the held-out campaign, and repeated this 32 times. The code verifies that no campaign appears in both the training and the test set.

The metric is the mean absolute error in percentage points, with each campaign weighted equally. The reference is a duration curve that uses only the number of days.

Slide 10 · LLM-based probabilistic classifier: input and output

About a minute and a half



The standard models receive coded columns. For the LLM, we described each measurement in text and passed it to an LLM-based probabilistic classifier. This model does not generate text. It receives a description and a fixed list of options and returns a probability for each option.

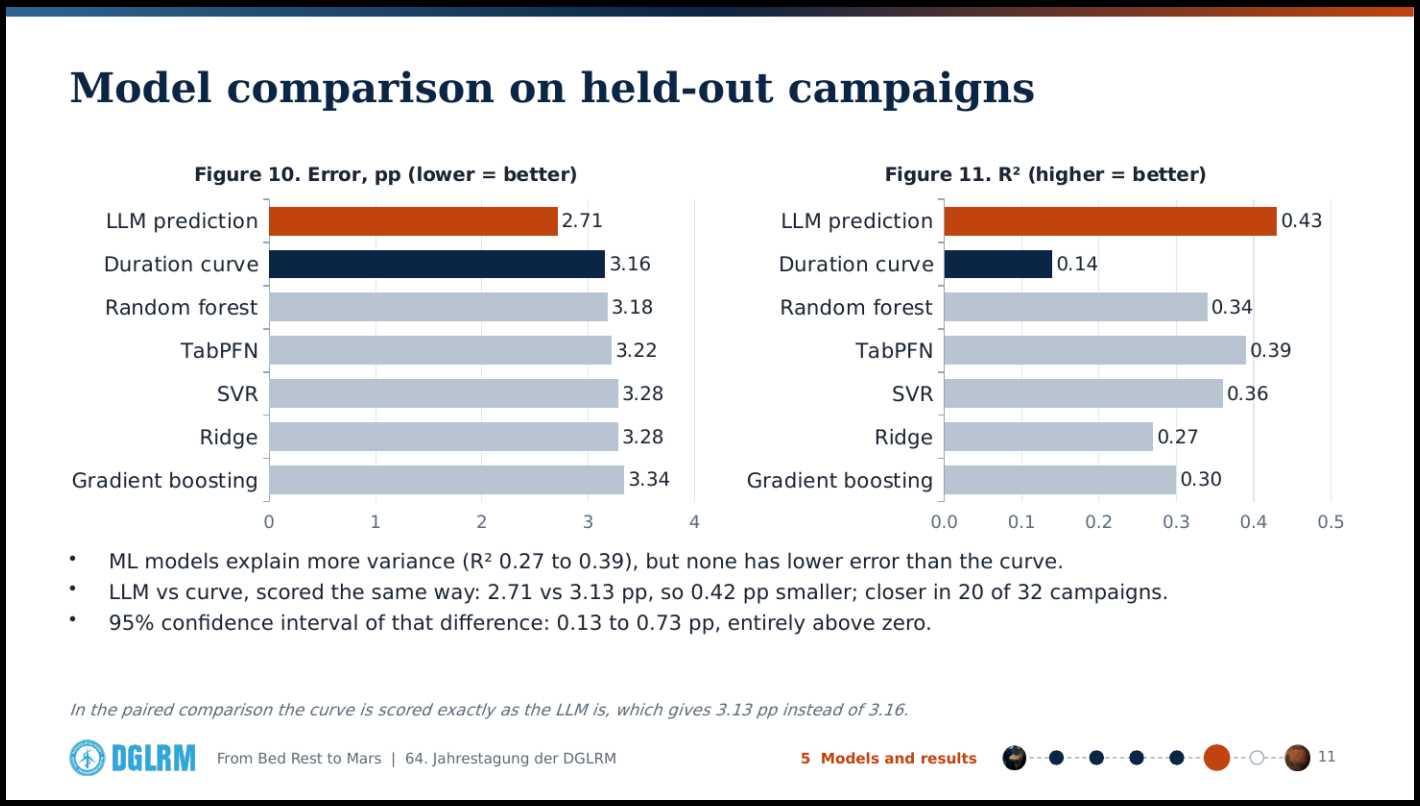
The left panel shows an actual request. It contains the participants, the protocol, the muscle, the imaging method and the target day. It also contains the duration curve fitted on the other 31 campaigns and the most similar rows from those campaigns. The model is asked in which of 19 ranges the change will fall.

The right panel shows the returned probabilities. We do not ask the model for a number. The prediction is the probability-weighted mean of the range midpoints, here −6.6 percent. The measured value was −9.4 percent. The error of this example equals the median error of the model.

The request never contains data from the held-out campaign or any paper, author or campaign name. This is checked automatically for every request. All 430 responses are stored, so every result can be reproduced without querying the model again.

Slide 11 · Model comparison on held-out campaigns

About a minute and a half



We compared four standard models, ridge regression, random forest, support vector regression and gradient boosting, each tuned within the training folds. We also tested TabPFN, a network pretrained for small tabular data.

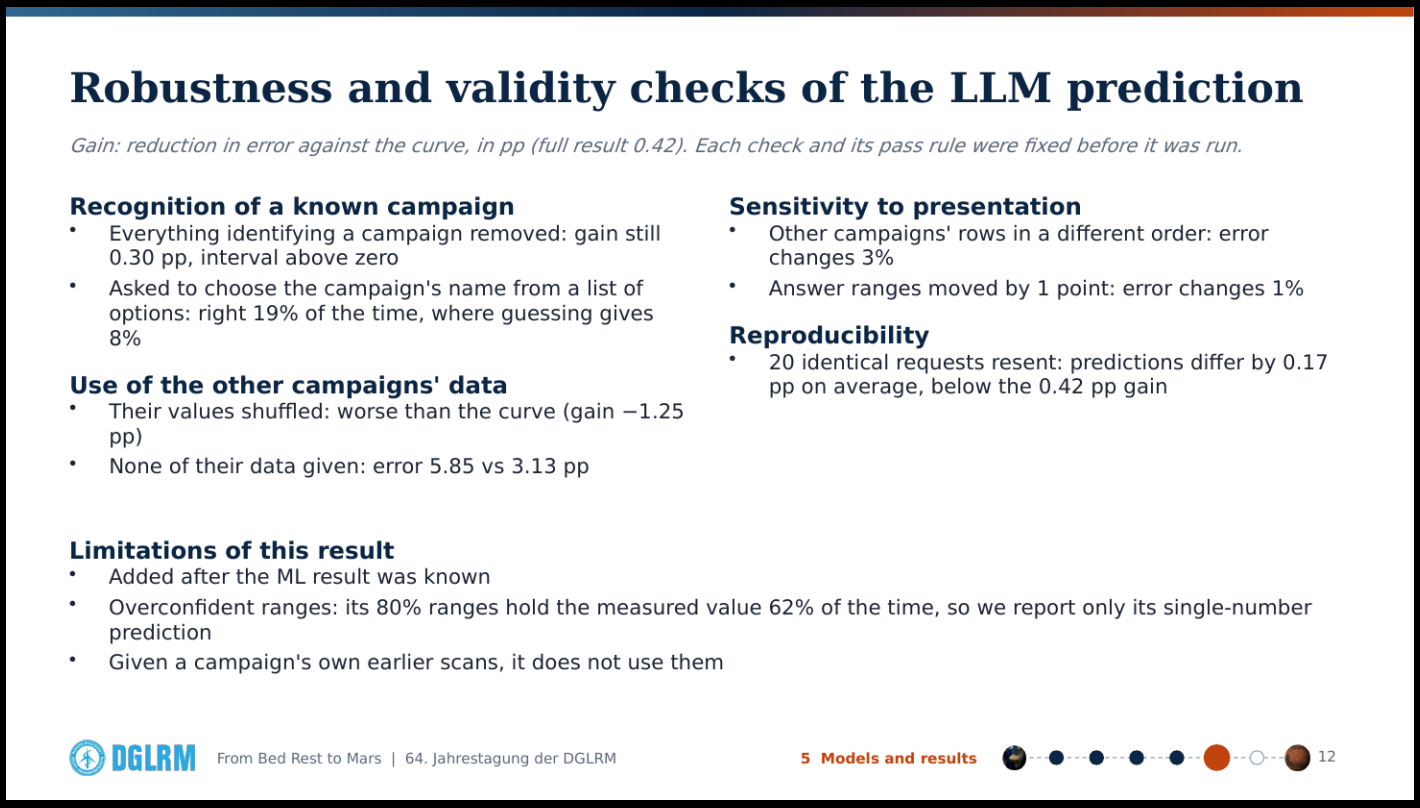
The left figure shows the error on the held-out campaigns. The duration curve has an error of 3.16 percentage points. The best machine learning model, the random forest, has 3.18. No machine learning model has a lower error than the curve. The right figure shows that these models explain two to three times more variance, because they learn the differences between muscles. Most of the remaining error is a shift that affects a whole campaign, for example through the scanner, the volunteers or the protocol. None of our inputs describes this shift. We had committed in advance to report this result regardless of the outcome.

The LLM has an error of 2.71 percentage points and the highest R^2, 0.43. In a paired comparison on the same rows, the error decreases from 3.13 to 2.71, a reduction of 0.42 percentage points. The 95 percent confidence interval is 0.13 to 0.73, so it excludes zero. The LLM is closer than the curve in 20 of 32 campaigns. No other model has an improvement with a confidence interval that excludes zero.

If asked about the target: we had set a 15 percent improvement in advance. The error improved by 13.3 percent and the probabilistic score by 17.7 percent.

Slide 12 · Robustness and validity checks of the LLM prediction

About a minute and a quarter



Each check and its pass criterion were defined and committed to the repository before the check was run.

First, recognition. The model may have seen the published studies during training. With all identifying details removed, the improvement is 0.30 percentage points, and the confidence interval still excludes zero. When asked to select the campaign name from a list, the model was correct in 19 percent of cases, against 8 percent by chance.

> *If asked about recognition: of the three campaigns that contribute most of the improvement, the model identified only one.*

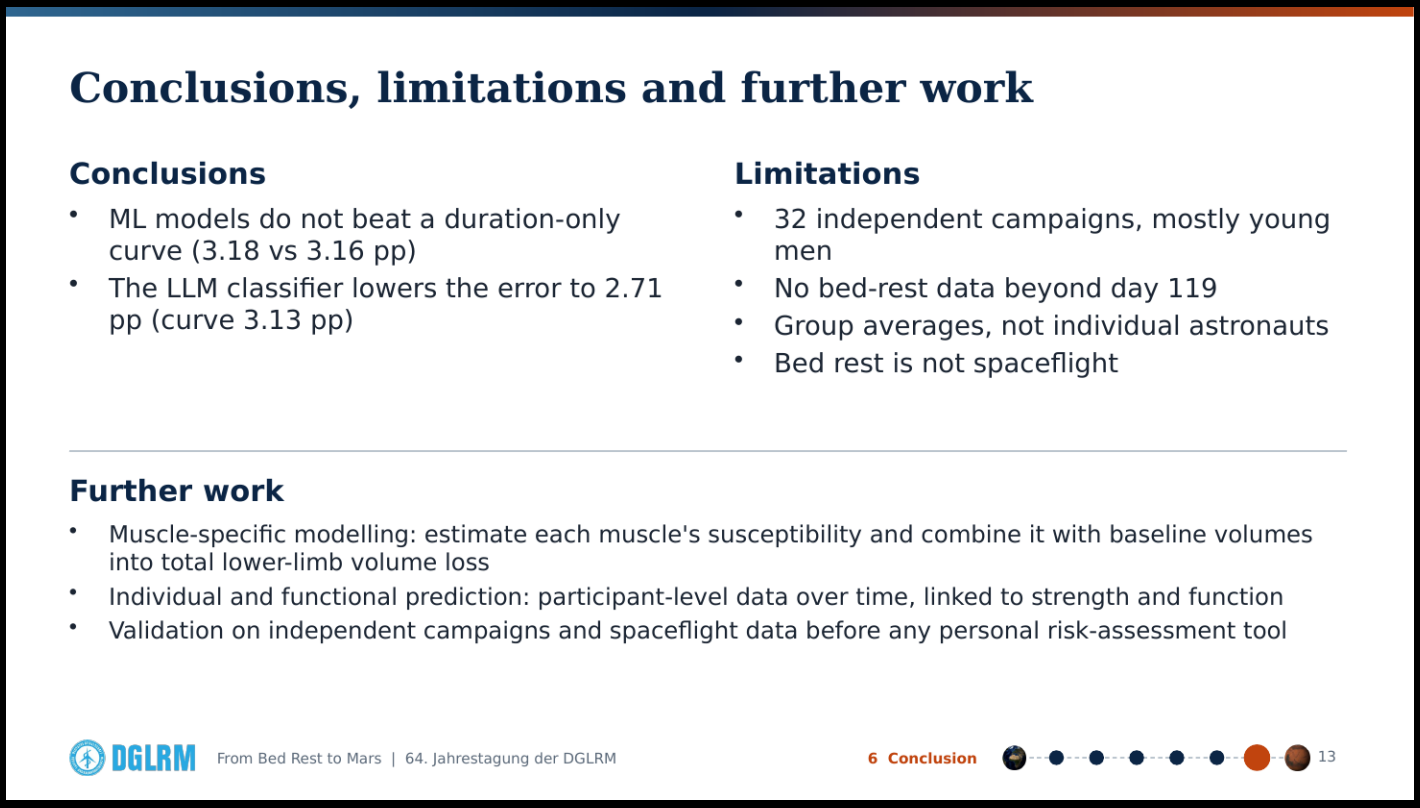
Second, use of the reference data. When the values of the other campaigns are shuffled, the model is 1.25 percentage points worse than the curve. Without reference data, the error rises to 5.85. The improvement therefore depends on these data.

Third, presentation. Changing the row order or shifting the range boundaries changes the error by 1 to 3 percent. Fourth, reproducibility. Repeating 20 requests changes the predictions by 0.17 percentage points on average, less than the 0.42 improvement.

There are three limitations. The model was added after the machine learning results were known. Its 80 percent ranges contain the measured value in only 62 percent of cases, so we report only point predictions. It also ignores a campaign's own earlier measurements.

Slide 13 · Conclusions, limitations and further work

About forty-five seconds



With 32 independent campaigns, standard machine learning models and TabPFN do not outperform a curve based on duration alone. The LLM-based classifier uses the description of each campaign and the data of the other campaigns, and it reduces the error from 3.13 to 2.71 percentage points.

The main limitations are the small number of campaigns, mostly with young men, and the lack of bed-rest data beyond day 119. Each row is a group mean. The model therefore predicts the average loss of a group, not the risk of an individual astronaut.

Further work has three steps. First, we will estimate the relative susceptibility of each muscle and combine it with baseline volumes to estimate the total volume loss of the lower limb. Second, we will use individual participant data over time and relate muscle loss to strength and function. Third, we will validate the model on independent campaigns and spaceflight data before developing an individual risk assessment.

If asked about the first step: the dataset contains 30 group time points with three or more muscles measured together, across 14 campaigns.

Slide 14 · Thank you for listening

Closing



Thank you for your attention. We are happy to take your questions.

Backup slides: 15 and 16 on the data and the search, 17 and 18 on part 1, 19 to 21 on part 2.

Slide 15 · Backup: dataset coverage and limitations

Backup: only if asked

Backup: dataset coverage and limitations

Did you miss studies?

- Four sources searched from 2013 on, plus 10 older or open-data studies (sources and queries on the next slide)
- Not covered: Embase (no access)

One campaign dominates the data

- MEDES 90-day: 40% of all rows, a quarter of the modelling data
- Scoring gives each campaign one vote; the re-run without it is still to do

Can MRI, CT, DXA and ultrasound be mixed?

- Each method gets its own term: DXA and ultrasound thickness show 5 to 6 pp less loss than CT; MRI volume is within 0.5 pp
- The MRI-only re-run is still to do

How reliable is the extraction?

- Every value is traced to its paper and page, table or figure
- 674 of 742 rows high confidence; 19 read off figures; no independent second extraction

If asked about sensitivity analyses: three planned analyses have not been run yet. These are the composite-first analysis, the analysis without the MEDES campaign, and the MRI-only analysis. Only the weighting check has been run, and it changed the results little.

Slide 16 · Backup: literature search sources and query

Backup: only if asked

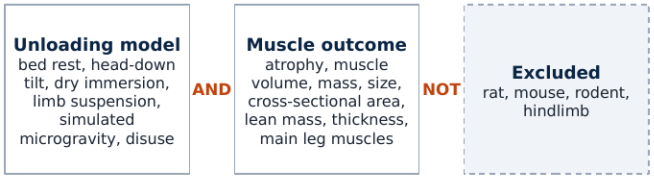
Backup: literature search sources and query

Table 3. Sources (searched 4 September 2026)

Source	Records
PubMed	1,412
Scopus	1,757
Web of Science Core Collection	1,876
NASA Technical Reports Server	686
Earlier studies and NASA open data	10
Total	**5,741**

- Journal databases: publications from 2013 onwards
- Older work enters through studies we held before the search (9) and one NASA open-data campaign
- NASA reports: five short keyword queries, e.g. "bed rest" "muscle volume"
- Not searched: Embase (no institutional access), Cochrane CENTRAL, trial registries, citation chasing

Figure 12. Query: two required blocks



Web of Science string, verbatim

```
TS=(("bed rest" OR bedrest OR "head-down tilt" OR "head down bed rest" OR HDBR
  OR antiorthostatic OR hypokinesia OR hypodynamia OR "dry immersion"
  OR "limb suspension" OR ULLS OR "simulated microgravity"
  OR "microgravity analog*" OR "spaceflight analog*" OR "mechanical unloading"
  OR "muscle unloading" OR disuse)
 AND
 (atroph* OR "muscle volume" OR "muscle mass" OR "muscle size"
  OR "cross-sectional area" OR PCSA OR "lean mass" OR "muscle thickness"
  OR "muscle wasting" OR deconditioning OR soleus OR gastrocnemius
  OR "triceps surae" OR quadriceps OR "vastus lateralis" OR "knee extensor"))
NOT TS=(rat OR rats OR mice OR mouse OR rodent OR hindlimb OR "hind limb")
```


All four search strings are in the appendix of the report and in docs/literature-review/search_log.md. PubMed uses the same two blocks with MeSH terms and a humans filter. Scopus and Web of Science exclude animal studies in the query.

The main weakness is the limit to publications from 2013 onwards. Only two of ten known modelling papers fall inside this window, and the search without the limit was not tested. Embase was not searched because we had no access. It indexes conference abstracts that the other databases miss.

Slide 17 · Backup: dataset expansion and composition

Backup: only if asked · Part 1 (Niloufar)

Backup: dataset expansion and composition

The abstract had 15 studies. Did adding 37 change the analysis after seeing data?

- Eligibility criteria were fixed before screening; the dataset was frozen (14 and 19 September) before the final models
- The curve shape was chosen by a rule declared before fitting

Slide 5 shows 180 days, but the model stops at day 119. Why?

- The 14 spaceflight rows were measured after landing, so they leave the model with the 264 recovery rows
- 8 of the 14 are one back muscle (lumbar multifidus)

Mostly young men: does this apply to women and older crews?

- Rows, not people: men only 567, mixed 113, women only 40 (35 of them WISE-2005), not reported 22
- Healthy young 692, middle-aged 8, older 42: the results describe young men

What exactly is a row's percentage change?

- Follow-up vs the same group's own baseline; negative means loss
- Recomputed from the printed values; where a paper prints the mean of individual changes, that value is kept and labelled
- Rows weighted by number of participants: only 8 of 32 campaigns report the spread of the change

The eligibility criteria in Table 2 of the report were defined before screening. Dataset version 1.0 was frozen on 14 September and version 1.1 on 19 September. All results use version 1.1.

The saturating curve is used unless another form is better by at least 4 AIC points.

The spaceflight rows remain in the dataset and are flagged. They were measured after landing, so the filter for measurements during unloading removes them.

Inverse-variance weighting would need the variance of the change, which most papers do not report. Comparing both weightings changed the duration effect by less than 1 percentage point.

Slide 18 · Backup: certainty of the muscle-group curves

Backup: only if asked · Part 1 (Niloufar)

Backup: certainty of the muscle-group curves

Is each line fitted to its own muscle?

- No. The day-60 values are fitted per muscle group (named muscles, 25 campaigns); the lines share one time course (τ = 60 days)
- Separate shapes cannot be estimated: six of nine groups rest on four campaigns or fewer

Why a saturating curve?

- Logarithmic, saturating and spline curves fit equally well (within 1.3 AIC points): the loss slows, but the data cannot say along which curve
- By day 119 the curve reaches 86% of its plateau, −17.3% (95% CI −20.9 to −13.7)

How certain are the day-60 values?

- Calf −15.8% (95% CI −18.9 to −12.7); front thigh −10.1% (−13.6 to −6.6); hip rotators −1.9% (−3.9 to 0.2)
- Calf vs shin, the ankle's opposing pair: 5.6 pp more loss (2.1 to 9.1), p = 0.003
- Hip rotators and outer hip: one campaign each

Is the calf result an artefact of how often it is scanned?

- The order holds after adjusting for duration, countermeasure group and imaging method
- Calf: 78 rows from 13 campaigns, the second best-covered group after the front thigh (86 rows, 22 campaigns)

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The dots at day 60 on slide 8 are the meta-regression estimates for each muscle group, with the intervals shown here. The lines use one shared time course scaled to these values. They show the assumed shape, not a separate fit for each muscle.

The intervals are 95 percent cluster-robust intervals, which account for several rows coming from the same campaign. The calf versus shin contrast is reported separately because it does not depend on the choice of reference muscle.

Slide 19 · Backup: significance and relevance of the LLM gain

Backup: only if asked

Backup: significance and relevance of the LLM gain

Is 0.42 pp worth anything?

- It is 13% lower error than the curve (3.13 to 2.71 pp)
- Small; we present it as a secondary finding

20 of 32 campaigns is close to a coin flip

- Counting wins only: p = 0.22 (sign test)*
- Using the size of each win: mean gain 0.42 pp, 95% CI 0.13 to 0.73

Does it rest on a few campaigns?

- 3 long MRI campaigns (NASA SPRINT, Berlin BBR2-2, MEDES LTBR) give 56% of the gain; without them 0.20 pp, CI −0.01 to 0.43*
- Only SPRINT is recognised, and with campaign details removed the LLM still wins there (3.79 vs 5.92 pp)

Did it read these papers in training? Was it added after the fact?

- No test can rule out training exposure; hiding campaign details and the naming test are the strongest checks available
- Added after the ML result, but every check and target was committed before the first answer

** Our calculation from the saved LLM predictions; not in the report.*

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The two numbers marked with an asterisk are our own calculation from results/forecast_predictions.csv. They are not in the report.

Without the three campaigns that contribute most of the improvement, the confidence interval includes zero. State this directly if asked.

Slide 20 · Backup: ML results and fairness of the comparison

Backup: only if asked

Backup: ML results and fairness of the comparison

Did the ML models get a fair chance?

- Tuned inside each training fold, grouped by campaign; TabPFN needs no tuning
- All models saw the same inputs, folds and scoring

Why not add age, sex or countermeasure type?

- 32 campaigns leave room for about 3 campaign-level inputs (about 10 campaigns per input)
- Age and sex are fixed within a campaign (271 of the 346 modelling rows men-only), so they act as a campaign label; 100 countermeasure rows spread over 9 types

Is the duration curve a straw man?

- Ridge regression uses days, muscle and every other input: 3.28 vs 3.16 pp
- Knowing the muscle raises R^2 but not the error on a new campaign

Would a nearest-neighbour method match the LLM?

- Not tested. If the gain comes from picking similar rows, this is the next control to run

The open point is the nearest-neighbour control. We cannot yet say whether a simple similarity method would perform as well as the LLM.

The counts of 271 men-only rows and 100 countermeasure rows refer to the 346 modelling rows. The report gives the same point for the 425 lower-limb rows before duplicate composites were removed: 347 men-only and 126 countermeasure rows.

Slide 21 · Backup: applicability to Mars mission planning

Backup: only if asked

Backup: applicability to Mars mission planning

A Mars transit is about 180 days; the data stop at 119

- Only one campaign reaches 119 days
- A 180-day estimate is planned from the duration curve only, with an uncertainty range, labelled as an assumption

Bed rest is not spaceflight

- Only 14 spaceflight rows, 8 of them one back muscle: too few to test transfer

Which countermeasure works? What about individuals?

- Countermeasure groups lose 3.7 pp less (95% CI 1.5 to 5.9), all types pooled
- Every row is a group average; nothing about individual risk

Can anyone reproduce an LLM result?

- Fixed model version; all 430 answers saved, so every number rebuilds without the model
- Identical requests differ by 0.17 pp on average

Within 5 to 119 days and at group level, the data support three statements for mission planning. Most of the loss occurs in the first weeks. The calf muscles need protection first. Groups with a countermeasure lose less muscle.